28 MAR 86
SUPERSEDING
MIL-I-83861/2(USAF)
29 May 1981

MILITARY SPECIFICATION SHEET
FILTER ELEMENTS, NONCLEANABLE, FLUID PRESSURE, HYDRAULIC LINE, 15 MICRON ABSOLUTE, TYPE II SYSTEMS

This specification sheet is approved for use by the Department of the Air Force and is available for use by all departments and agencies of the Department of the Defense.

The complete requirements for acquiring the filter element described herein shall consist of this specification and the latest issue of Specification MIL-F-83861(USAF).

REQUIREMENTS:

Bubble point. The initial and burst out bubble point values shall meet the values established by the activity responsible for qualification and shall be determined as specified in 4.7.1.1 of general specification.

Filtration. The filter element shall remove 100 percent of all contaminants over 15 microns and 99 percent by weight of add contaminants.

Cold start. The filter element shall withstand 3000 psi differential pressure across the element for the flow cycles and fluid temperature as specified in 4.7.3 of general specification.

Element pressure drop. The pressure drop through the clean element at rated flow shall not exceed 12 psid. Test fluid shall be maintained at 100 $\pm$ 10°F.

Element dirt capacity. The filter element shall meet the dirt capacity specified in Table I.

Element collapse. The element shall withstand 4500 psi differential pressure.

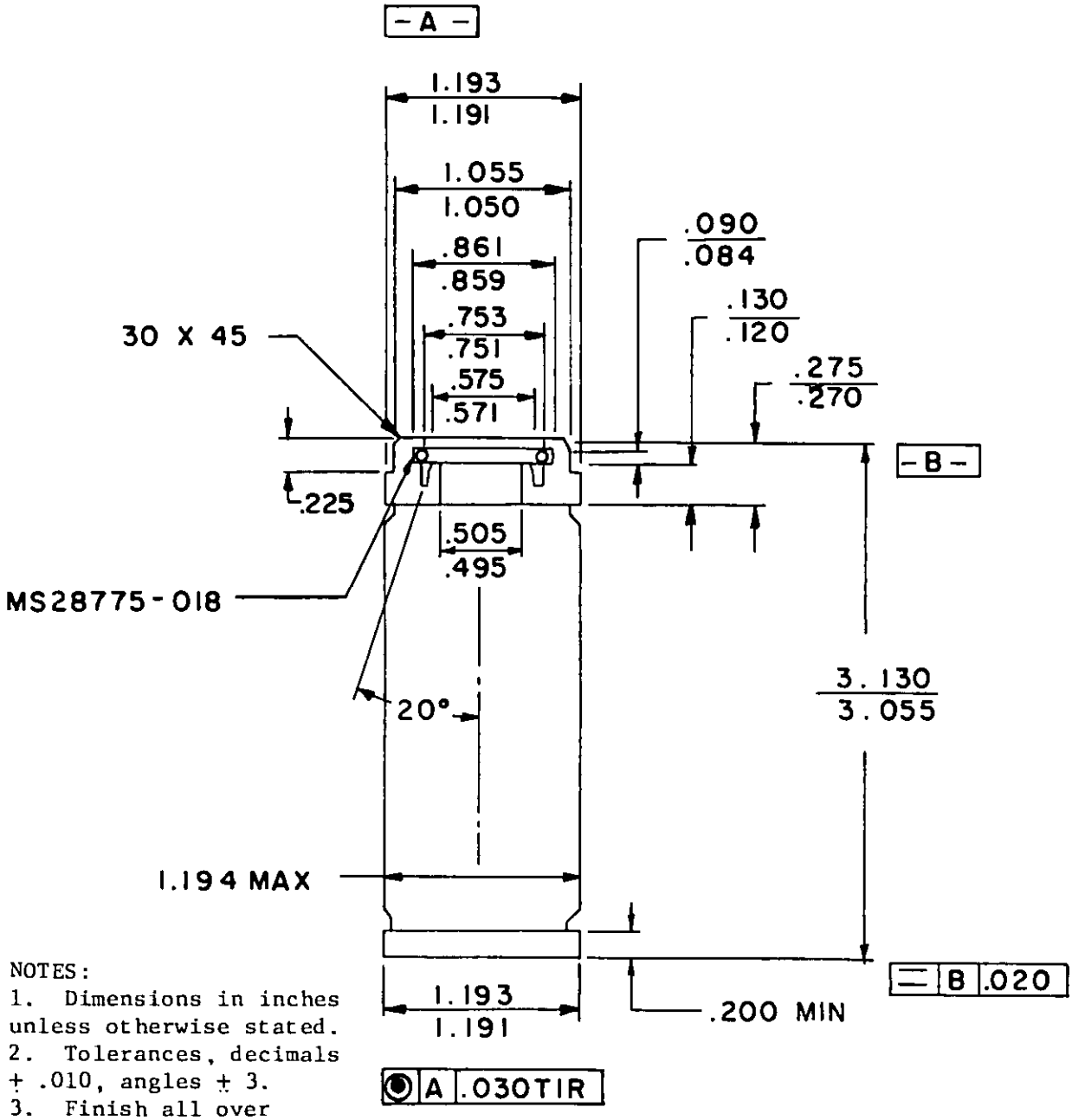
Description. The element shall be shown in Figure 1 and as specified herein.

Flow fatigue. The filter elements shall withstand the flow fatigue cycles specified at 300 cpm:

Cycles	Pressure drop (PSID)
1,120,000	20
112,000	50
12,500	100
400	1500
100	3000

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil - Downloaded: 2026-10-25T09:51Z
Check the source to verify that this is the current version before use.



NOTES:
1. Dimensions in inches unless otherwise stated.
2. Tolerances, decimals ± .010, angles ± 3.
3. Finish all over 100RMS except as noted.
4. Material shall be in accordance with the procurement specification.

FIGURE 1. Filter element, 15 micron absolute.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:51Z
Check the source to verify that this is the current version before use.

Media migration. Element migration when tested in accordance with 4.7.8.3.2 of general specification shall not exceed 0.0040 grams and there shall be no migration identifiable as coming from the filter media.

Maximum particle passed. The maximum particle passed shall be no greater than 15 microns.

Part number:
The part number consists of the basic number of this specification sheet and a dash number taken from Table I. (see page 3).

Example: M83861/2-1

Intended use:
The filter element covered by this specification sheet is intended for use in model C130 aircraft operating with hydraulic fluid conforming to MIL-H-5606 at nominal operating pressures of 3000 psi.

TABLE I. Dash number and operating characteristics.

Dash no.	Flow rate (gms)	Degree of filtration add rate	Maximum particle add rate	Pressure buildup	
		F-9 Beads (gms)	F-9 Beads (gms)	Add rate AC dust (gms)	Minimum AC dust to 90 psid (gms)
-1	12	0.30	0.30	0.4	1.5

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodian:
Air Force-99

Preparing Activity:
Air Force-71

Project Number:
1650-F130

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:51Z
Check the source to verify that this is the current version before use.